

Explainable Machine Learning for Post-Operative Patient Outcome Prediction Using DALEX

Dhrubajyoti Ghosh*, Tapas Si†, *Member, IEEE*, Samaresh Maiti*, Sarbajit Manna*
*Dept. of Computer Science & Electronics, Ramakrishna Mission Vidyamandira, Belur Math, Howrah

West Bengal-711202, India

Email: dhrubajyotighosh379@gmail.com, sarbajitonline@gmail.com

ORCID: 0000-0001-7669-9311, maitisamaresh@gmail.com

† UEMJ Advanced Machine Learning Research Group

AI Innovation Lab, Dept. of Computer Science & Engineering

University of Engineering & Management Jaipur, Rajasthan - 303807, India

Email: tapasi.aiml@gmail.com, ORCID: 0000-0001-8267-0304

Abstract—Post-operative complications are still a big problem in surgery. They lead to more illness, longer hospital stays and higher healthcare costs. The prediction of such complications during the early stages can be helpful in taking appropriate decisions. In this study, Four ML classifiers— Decision Tree (DT), Extreme Gradient Boosting (XGBoost), Multi-Layer Perceptron (MLP), and Stacking Classifier (SC) — is used to predict treatment outcomes. We use Synthetic Minority Oversampling Technique (SMOTE) to balance the classes of the dataset. Stratified K-Fold cross validation technique is also used to validate the model properly. The models' performances are measured by accuracy, recall, precision, F_1 -Score, Geometric Mean (GM) and False Positive Rate (FPR). The Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) method from Multi-Criteria Decision Making (MCDM) identifies XGBoost as the top-performing model, achieving mean accuracy of 82.81%, precision of 84.55%, recall of 82.81%, F_1 of 82.56%, GM of 86.98%, and FPR of 8.61%. To improve the model's interpretability, Descriptive ML Explanations (DALEX) is used to find important features affecting the model's predictions. The study contributes to the development of intelligent decision-support systems, helping in the reduction of post-operative complications.

Index Terms—Post-operative outcome, Machine Learning, SMOTE, TOPSIS, DALEX

I. INTRODUCTION

Post-operative patient management is a critical phase in healthcare because patients face a high risk of developing severe complications which include delirium, infection, organ failure, and a long recovery time. Good predictions of outcomes provide useful information for making good clinical decisions and decreasing risks of complications. ML is now a very useful tool for using large amounts of clinical data to predict surgical outcomes more accurately than previously through traditional statistical methods. Several studies have explored the application of ML techniques in predicting post-operative complications. Bishara et al. [1] developed models using neural networks and XGBoost to predict post-operative delirium, achieving an AUC of 0.851, outperforming traditional methods. However, the model is limited to preoperative data. Racine et al. [2] also used ML techniques like logistic regression (LR), neural networks, and gradient boosting to

predict problems in older patients with a moderate AUC (0.62–0.71) value. Zeng et al. [3] used decision tree models to predict problems after liver surgery, achieving high accuracy (92.45%), though with a relatively small dataset. In the subsequent studies, the complications of acute kidney injury and infections have also been addressed. In the study by Li et al. [4], the implementation of the LR-based ML model for the prediction of severe acute kidney injury in patients who have undergone heart transplantation is discussed with an AUC of 0.821. However, the generalizability of the model is limited due to the small sample size. Heffernan et al. [5] analyzed a large dataset of over 600,000 patients and found that XGBoost outperformed other models in predicting post-operative infections. Yasin and other people [6] used XGBoost with SMOTE on the patient records of 580 people to figure out how long patients with tuberculosis spondylitis would stay in the hospital. They were able to predict this well with an AUC of 0.86. Göktürk et al. [7] used LightGBM to predict if patients would have bleeding in their brain after surgery. They were able to predict this well with an AUC of 0.745 and they found some important biomarkers. Nopour [8] developed predictive models for breast cancer recurrence using multiple ML algorithms, where RF (RF) achieved the best performance (AUC 0.919, accuracy 94%), supported by SHAP-based interpretability. However, most of the existing literature on the prediction of post-operative face several common limitations. Firstly, The studies depend on either small databases or single-center data which limits their ability to produce results that apply to various patient groups. Secondly, there is also the problem of imbalance and low diversity of the samples used for prediction in these studies and limited clinical validation, and lack of interpretability. One of the biggest issues is that many of the ML models function as 'black box' models, meaning that doctors cannot easily interpret the results and therefore do not have confidence in their predictions. To address these concerns, XAI methods like DALEX [9], [10] have been introduced. In this study, we apply ML classifiers DT, SC, XGBoost, MLP –to a UCI Post Operative patient dataset and evaluate them using multiple matrices (accuracy,

precision, recall, F_1 , GM) We use SMOTE [11] to balance the imbalance dataset. We use TOPSIS [12] from MCDM [13] to select the best model based on all performance metrics and integrate DALEX to visualize feature importance and Breakdown plot for all classes. A Stratified 10-fold cross validation approach improves model generalization and decreases overfitting. By integrating robust predictive modeling with comprehensive evaluation and explainable AI methods this work aims to enhance early risk detection and support more reliable, data-driven decision-making in post-operative patient care. DALEX-based interpretation of different ML models in the prediction for Post-Operative patient outcome is the key novelty of this current research work.

II. RELATED WORKS

Bishara et al. [1] aimed to create ML models such as neural networks and XGBoost that predict the risk of postoperative delirium (POD) using preoperative electronic health record (EHR) data. The research was based on a large amount of retrospective data from two large hospitals, including 24,885 surgical patients over the age of 18. The data set consisted of 115 preoperative variables that were used as predictors for each of the models and was compared to traditional LR and the standard delirium risk assessment tool, while the prediction model developed using XGBoost had an area under the curve (AUC) of 0.851, making it superior to other developed models. It must be noted that it was based only on pre-operative EHR data and left out intra-operative physiological variables, which could improve the accuracy of predictions further.

Racine et al. [2] used ML methods to predict POD in 560 patients over the age of 70 who were undergoing major elective surgeries. They collected 71 different clinical indicators, including various demographic data, cognitive test scores and medical information obtained from the patients medical records. The team used three ML algorithms such as LR, Neural networks and Gradient Boosting. Their models achieved AUC scores between 0.62 and 0.71, showing a moderate level of accuracy in the prediction. According to the study, the ML models performed equally as well as traditional statistical models in the prediction of POD. However, the study noted that the prediction of POD remains a challenge, mainly due to the complexity of finding the right clinical factors, as well as the limited amount of available data.

Zeng et al. [3] developed a ML model to predict complications for liver resection patients. They collected data from 175 people who had liver surgery for cancer and examined 13 clinical factors, including BMI, operation duration, incision length, and basic patient information. They compared the performance of five different ML algorithms such as LR, DT (C5.0 and CART), SVMs(SVM), and RF. They found that the C5.0 decision tree algorithm performed the best, with an AUC value of 0.91 and an accuracy of 92.45%, which showed how well ML methods could forecast surgical complications that occur during the surgery procedure. However, the number of samples used in the study is relatively low, i.e., 175 patients, and the results may not be generalized to larger populations.

In Li et al.'s study [4], they created a predictive ML model to predict stage 3 acute kidney injury (AKI), which is one of the most serious complications that can occur after heart transplantation. The model was developed using data from 107 patients who had received heart transplants between 2018 and 2020 from a single hospital. The predictive model was developed through a regularized LR approach. The features included are creatinine levels, glomerular filtration rate, dimensions, and coagulation indicators. The model performed better than the previous Cleveland Clinic clinical predictive model with an AUC of 0.821. Overall, these results generally indicate that use of ML will improve the earliest detection of important multiple complications. Although the study also had its own limitations. The number of patients was limited, and all the patients were from one center, so they were unable to test the model's effectiveness.

In a study by Heffernan et al. [5] they examined the capability of different models of ML to predict the development of post-operative infections of undergoing surgery patients using a huge dataset from the National Surgical Quality Improvement Program (NSQIP) that included 624,625 patients. They use several algorithms such as XGBoost, RF, LR, and K-Nearest Neighbors (KNN). XGBoost algorithm had the highest predictive performance, with an area under the curve (AUC) value of 0.85, while LR had an AUC of 0.82. The study highlighted that ML models differ in their ability to predict patient outcomes. So the choice of models is very important.

The study done by Yasin et al. In [6] looked at how to predict if patients with tuberculosis spondylitis will have a hospital stay. They analyzed data from 580 patient records to conduct their study. The healthcare data presents challenges because it contains complex information which consists of multiple components. In this study, XGBoost algorithm generated positive outcomes, with an AUC score close to 0.86. Also, to adjust the imbalance the dataset, they use the SMOTE. But even with these good points, the study also had some limitations, as it was based on data collected from a single center and on only one disease.

Göktürk et al. [7] suggested the use of ML to predict postoperative intracranial hemorrhage in neurosurgery. In this study, they use a small dataset of 118 patients. They use LightGBM algorithm, with an AUC score of 0.745. The role of biomarkers such as C-reactive protein and platelet was also highlighted in this research. They also used SHAP and LIME to understand how the model works. The results are affected because the number of patients is limited, and all the patients are from a single source.

Nopour et al. [8] developed a predictive models using a dataset of 1156 patients from different clinical centers. The dataset was on one-year recurrence in post-operative breast cancer patients. In this study, they used Neural networks, RF, XGBoost, SVM, and other ML and deep learning algorithms. The accuracy of 94% and AUC 0.919 were obtained by the RF model, which was higher than other models. The study also used SHAP-based XAI analysis to identify several important prognostic indicators (such as tumor grade, HER2 status,

lymph node involvement), which made the results easier to understand. Since the study made accurate predictions, the study was based on historical data, and its accuracy was not tested on different populations.

In Zhou et al. [14] study, they introduced a multicenter ML framework which predicts short-term outcomes for patients with aneurysmal subarachnoid hemorrhage. The study combined clinical severity scores with inflammatory biomarkers. They used a large dataset with 1120 patients to develop the model. They also used another set with 326 patients to validate the model combining clinical severity scores with inflammatory biomarkers. Multiple models were implemented and tested, and the highest performance was obtained by Gradient Boosting Model (AUC = 0.895 in internal validation and 0.864 in external validation). SHAP and LIME are both used in the study to interpret the results and obtain a global and local understanding of the model's predictions. This dual XAI approach made it easier to understand clinical data and helped put a web-based decision support tool into use.

III. MATERIALS & METHODS

A. Dataset Description

The dataset we are using for our study is called the Post-Operative Patient Dataset and was obtained from the UCI ML Repository [15]. The dataset includes medical records of patients who have undergone surgical procedures and aims to predict their post-operative recovery results. The dataset consists of 90 patient instances, each described by eight input features and one target variable. This dataset has 64 samples of class 'A' (discharge), whereas 'S' (general ward) and 'I' (ICU) have 24 and 2 samples respectively. After using SMOTE all classes have 64 samples which balance the dataset. The input features are : L-CORE (patient's core body temperature), L-SURF (surface body temperature), L-O2 (oxygen saturation level), L-BP (blood pressure level), SURF-STBL (surface temperature stability), CORE-STBL (core temperature stability), BP-STBL (blood pressure stability) and COMFORT (patient comfort level). The target variable, ADM-DECS (admission decision), indicates the post-operative outcome category of the patient. It is a multi-class classification problem, where patients are classified into one of three categories: I (patient sent to Intensive Care Unit), S (patient sent to general hospital floor) and A (patient discharged home). Most features like L-CORE, L-SURF, L-O2, L-BP, SURF-STBL, CORE-STBL, BP-STBL are categorical (nominal/ordinal) because they contain values like high, mid, low, stable, unstable, etc. However, COMFORT is treated as a numerical feature. The dataset has been widely used for testing the performance of classification algorithms for the prediction of patient outcome based on post-operative patient physiological status.

B. Data Preprocessing

Different types of preprocessing techniques and methods were applied to improve data quality and enhance model performance.

1) *Replacing Missing Values:* There are missing data in the COMFORT feature of the dataset, which we converted to numbers, and we filled in the blanks using the median imputation method. We chose the median because the value is not influenced by outliers and maintains the same distribution.

2) *Encoding of Categorical Variables:* XGBoost, MLP cannot process categorical values directly. They require numerical input. One-hot encoding changed the input features from categorical values to binary numbers. Using encoding method, which turned each class ('A', 'S', 'I') into a number.

3) *Handling Class Imbalance:* The Post-Operative Patient dataset has a class imbalance problem. This dataset has 64 samples of class 'A' (discharge), whereas 'S' (general ward) and 'I' (ICU) have 24 and 2 samples respectively. This imbalance makes the models focus on the majority class, which makes it harder for them to correctly predict cases in the minority class, especially important ones like ICU admissions. To fix this problem, the SMOTE method is used. SMOTE is a good choice for this dataset because it makes synthetic samples for minority classes while keeping the data's categorical nature. When we use SMOTE, all classes have 64 samples, which helps the models learn patterns from all classes better. This method helps to improve the performance of models, balancing the classes. That enhances evaluation metrics such as recall, F_1 -Score, and GM.

C. Stratified K-fold Cross Validation

A resampling technique, known as the stratified K-fold cross validation, involves partitioning the data into K folds, ensuring that the class distribution in each fold is the same, for the evaluation of the performance of ML algorithms. One fold is chosen as the validation set for each iteration, and the other folds are used for training. Until every fold has been used for testing, this procedure is repeated. Stratified sampling offers a more precise and reliable estimate of model performance by maintaining class proportions, particularly for unbalanced datasets.

D. Models

We used XGBoost, DT, MLP and SC to predict post-operative patient outcomes based on physiological and clinical parameters. The ML models we used in this work are described below.

1) *XGBoost:* XGBoost is an efficient ensemble learning method based on the principle of building a sequence of decision trees using the gradient boosting method, where each decision tree corrects the mistakes made by the previous one, thus enhancing the overall prediction accuracy. It also uses regularization, tree pruning, and parallel processing, making it efficient for handling structured data.

2) *Decision Tree:* Decision Tree is a supervised learning algorithm of the rule-based type and is used for classification and regression tasks. It recursively partitions the data into

subsets based on the values of the features and creates a tree-like structure where the internal nodes represent the decisions and the leaves represent the outcomes. It is a popular baseline algorithm due to its simplicity and ability to work well with both nominal and numerical data.

3) *MLP Classifier*: MLP Classifier is a type of artificial neural network consisting of multiple layers of neurons. MLP uses activation functions and backpropagation to learn complex non-linear relationships. The MLP model works best with datasets whose patterns are not linearly separable.

4) *Stacking Classifier*: SC is a ML method which combines several base models to make predictions more accurate. In this study, KNN, Naive Bayes, and RF are used as base models. Then, their predictions are sent to a meta-classifier, which in this case is Gradient Boosting. This meta-classifier learns how to best combine the outputs of the base learners. This method uses the advantages of various algorithms. For example, RF makes reliable tree-based predictions, Naive Bayes models probabilistic relationships well, and KNN finds patterns based on similarity. The Gradient Boosting meta-model makes these predictions more valuable by fixing mistakes and improving generalization. This make the model more accurate.

E. Evaluation Metrics

We used five performance metrics for this study: Accuracy, Precision, Recall, F_1 -Score, GM and FPR [16], [17].

IV. EXPERIMENTAL SETUP

A. Selecting Best Parameters

Hyperparameter tuning was performed using the Random Search method [18] with a 10-fold Stratified cross-validation strategy. A set of parameters for each model was defined. The hyperparameters were sampled randomly 20 times. In each case, the model was trained with different folds with similar class distribution. The model's performance was evaluated with respect to accuracy. The hyperparameters that provided the highest average accuracy over all folds were used to build the model. Optimised hyperparameter combinations are shown in Table I.

TABLE I
BEST HYPER PARAMETERS FOR MODELS

Model	Best Parameters
XGBoost	subsample=1.0, reg_lambda=1, reg_alpha=0.1, n_estimators=300, min_child_weight=1, Maximum Depth=5, Learning Rate=0.05, Gamma=0, colsample_bytree=0.6
MLP	Solver=adam, Learning Rate_init=0.01, Learning Rate=adaptive, Hidden Layer Size=(50), Alpha=0.001, Activation=relu
DT	min_samples_split=5, min_samples_leaf=1, max_features=None, max_depth=20, criterion=log_loss
SC	mlp_hidden_layer_sizes=(100), mlp_alpha=0.001, mlp_activation=relu, final_estimator_C=10, dt_max_depth=5

B. System Configuration

Model training and evaluation were conducted using the free version of Google Colaboratory. The environment was

equipped with the following specifications: Platform: Google Colaboratory, Processor: Intel® Xeon® CPU @ 2.20GHz, RAM: 12.00 GB, and Operating System: Linux (Debian-based).

V. RESULTS & DISCUSSION

In this study, experiments were conducted using the Post-Operative Patient Dataset obtained from the UCI ML Repository to evaluate the performance of various ML models, including XGBoost, DT, MLP, and SC. Several performance metrics were used to test the effectiveness of the models on this data. DALEX was implemented to visualize feature importance and Breakdown plot for all classes.

A. Training Accuracy and CPU Time

Training accuracies and CPU times are shown in Table II. Models with their best parameters are tested on the training data for each of the 10 folds, and their accuracies and CPU times are averaged. All of the models achieved very high accuracies(90-98)%. MLP Classifier and SC both achieved the highest mean accuracy of 97%, with CPU time of 0.093 sec and 1.9533 sec respectively. Where MLP Classifier is more computationally efficient than SC. XGBoost achieved training mean accuracy of 96% with CPU time of 0.177 seconds. DT get 93% accuracy. DT takes a very low computation time of 0.013 seconds. So DT proves to be the most effective model computationally, with a CPU time of 0.013 seconds.

TABLE II
TRAINING ACCURACIES AND CPU TIME

Model Name	Training Accuracy (%)		Mean CPU Time (s)
	Mean	Std. Dev.	
XGBoost	96.99	0.005	0.1772
MLP	97.00	0.006	0.093
SC	97.00	0.006	1.9533
DT	93.00	0.010	0.013

B. Performance Evaluation

The testing performance of the models are shown in the table below. In each CV split, performance metrics were computed on the test data, and their mean and standard deviation are detailed in the table.

In our study, XGBoost performed better than all models with the best accuracy (82.81%), precision (84.55%), recall (82.81%), and F_1 -Score (82.56%) and GM (86.98%) that indicate a balanced performance across all classes. It also gets the lowest standard deviation among the models, showing that the model is stable and consistent across all folds. The SC also showed promising results since it obtained a mean accuracy of 82.26%, precision of 83.94%, recall of 82.26%, F_1 -Score of 81.99%, and a GM of 86.59%. This again proves the concept of combining models for better results. The MLP classifier had similar results, with an average accuracy of 81.76%, a precision of 83.42%, a recall of 81.76%, an F_1 -Score of 81.45%, and a GM of 86.20%. However, the DT classifier obtained the lowest results compared to all the classifiers

TABLE III
MAXIMUM, MINIMUM, MEAN AND STANDARD DEVIATION OF PERFORMANCE METRICS OVER 10-FOLDS

Models	Metric	Acc.	Prec.	Recall	F_1 -Score	GM	FPR	TOPSIS Score	TOPSIS Rank
XGBoost	Best	89.47	92.10	89.47	89.17	92.26	4.85	1.0000	1
	Worst	73.68	74.18	73.68	73.68	79.86	13.42		
	Mean	82.81	84.55	82.81	82.56	86.98	8.61		
	Std. Dev.	4.06	5.14	4.06	3.94	3.18	2.23		
MLP	Best	89.47	92.10	89.47	89.17	92.26	4.85	0.9275	2
	Worst	73.68	74.18	73.68	73.68	79.86	13.42		
	Mean	81.76	83.42	81.76	81.45	86.20	8.61		
	Std. Dev.	4.24	4.88	4.24	4.34	3.34	2.23		
DT	Best	89.47	91.81	89.47	89.07	91.64	6.14	0.0000	4
	Worst	68.42	67.21	68.42	67.00	75.87	15.85		
	Mean	78.65	81.43	78.65	78.05	83.85	10.54		
	Std. Dev.	7.48	8.88	7.48	7.67	5.60	3.27		
SC	Best	89.47	92.10	89.47	89.17	92.26	5.12	0.6657	3
	Worst	73.68	73.55	73.68	73.36	79.57	13.46		
	Mean	82.26	83.94	82.26	81.99	86.59	10.13		
	Std. Dev.	5.16	5.58	5.16	5.21	4.00	2.47		

since it obtained a mean accuracy of 78.65%, precision of 81.43%, recall of 78.65%, and an F_1 -Score of 78.05%. The high standard deviation also proves that the DT classifier was not consistent in all the folds since it obtained higher standard deviation values compared to the others. The low GM of 83.85% also proves that it was not able to handle class imbalance well. Overall, ensemble-based methods, especially XGBoost and the SC, were better at making predictions and more reliable than single models. According to TOPSIS rank, **XGBoost** model is the top-performing model. These methods are also more applicable for predicting post-operative patient outcomes since accurate and balanced classification is required for such predictions. Zeng et al. [3] found that the C5.0 decision tree algorithm performed the best, with an AUC value of 0.91 and an accuracy of 0.924. Nopour et al. [8] obtained accuracy of 0.94 and AUC 0.919 using the RF model whereas we achieved 0.821 accuracy and 0.869 GM using XGBoost model. Overall, the performance of all models is closely comparable, which shows that the feature set, and preprocessing steps are effective for the given classification task.

C. Explainability Using DALEX

1) *XGBoost*: Fig. 1 indicates that in the XGBoost model, features such as L_CORE_mid, L_SURF_low, SURF_STBL_stable, and COMFORT have a positive contribution towards the prediction outcome. Among these features, features such as L_CORE_mid and COMFORT have a strong impact on the prediction outcome, as they have a significant impact on increasing the prediction score.

2) *MLP*: Fig. 2 shows that in the MLP model, COMFORT and L_O2_excellent are the most important factors that greatly raise the chance of making a correct prediction. L_SURF_low and SURF_STBL_stable are two other features that have a clear positive effect. Other variables, such as CORE_STBL

and BP_STBL, have smaller effects. This is due to the ability of the MLP model to handle non-linear relationships.

3) *DT*: As presented in Fig. 3, BP_STBL_stable is the most dominant feature in the Decision Tree model, as it greatly increases the prediction outcome. COMFORT and L_CORE_mid also have a positive impact on the prediction outcome, but not as much. On the other hand, features like SURF_STBL_stable and L_SURF_low make the prediction worse, which lowers the final output. The model depends on a small number of important features, which makes the contributions less balanced than those of ensemble models.

4) *SC*: The Stacking Classifier's breakdown analysis is shown in Fig. 4, which shows how different features affect the final prediction for a particular instance. Every feature either raises or lowers the prediction score, with the baseline (intercept) beginning at roughly 0.369. Among the features, L_BP_mid and SURF_STBL_stable have the most significant positive contributions, increasing the prediction by approximately +0.152 and +0.128, respectively. Like other features, L_O2_excellent and L_SURF_low also help the model predict better results. Suggesting that stable physiological conditions and favorable oxygen levels have a significant impact on the model's ability to predict better results. The feature 'COMFORT' makes a small but positive contribution. But some features make the prediction worse. Particularly, L_O2_good has the biggest negative contribution (−0.154). CORE_STBL_unstable and CORE_STBL_stable, which both have a small negative impact on the prediction. After looking at all the effects of all the features, including "other factors," it can be said that the final prediction probability is about 0.787.

VI. CONCLUSION

According to the result, ensemble-based models, specifically XGBoost, outperforms individual models such as MLP, DT, and SC in predicting post-operative patient outcomes. These

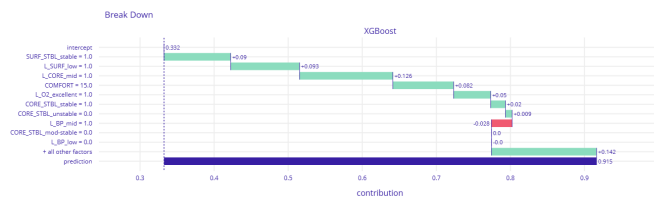


Fig. 1. DALEX Breakdown Plot for XGBoost Classifier.

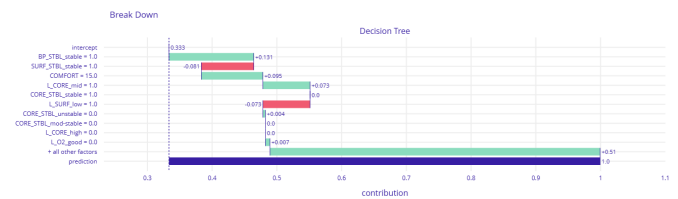


Fig. 3. DALEX Breakdown Plot for Decision Tree Classifier.

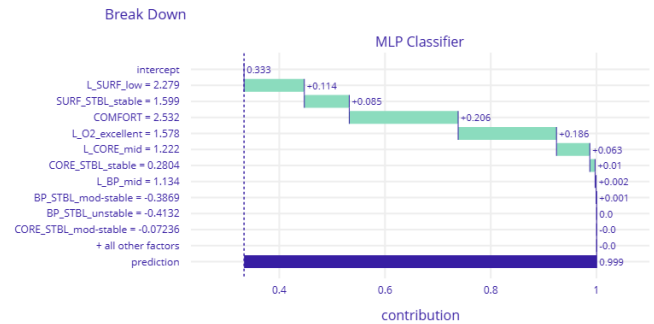


Fig. 2. DALEX Breakdown Plot for MLP Classifier.

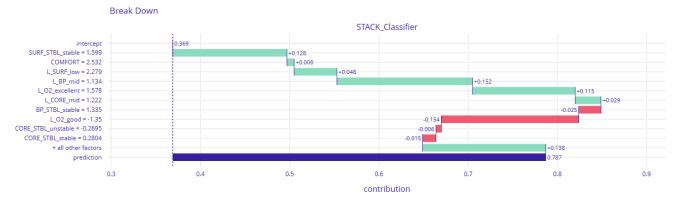


Fig. 4. DALEX Breakdown Plot for Stacking Classifier.

models have better accuracy, F_1 -Score, and GM values, which means they are better at handling the challenge of class imbalance. The performance of the DT model revealed that it was not able to perform or provide accurate results. The use of explainability techniques such as DALEX has made it easier to understand the predictions made by the models. These tools give valuable insights into the predictions and the contributions of each parameter to the overall prediction results, which makes them useful for making clinical decisions. The future work could be to test the model on large and diverse datasets to enhance the model's performance and make it robust. You can also use other advanced performance matrices like AUC, Cohen's Kappa, and others to see how well these models work for decision support systems in medicine.

REFERENCES

- [1] A. Bishara, C. Chiu, E. L. Whitlock, V. C. Douglas, S. Lee, A. J. Butte, J. M. Leung, and A. L. Donovan, "Postoperative delirium prediction using machine learning models and preoperative electronic health record data," *BMC anesthesiology*, vol. 22, no. 1, p. 8, 2022.
- [2] A. M. Racine, D. Tommet, M. L. D'Aquila, T. G. Fong, Y. Gou, P. A. Tabloski, E. D. Metzger, T. T. Hsieh, E. M. Schmitt, S. M. Vasunilashorn *et al.*, "Machine learning to develop and internally validate a predictive model for post-operative delirium in a prospective, observational clinical cohort study of older surgical patients," *Journal of general internal medicine*, vol. 36, no. 2, pp. 265–273, 2021.
- [3] S. Zeng, L. Li, Y. Hu, L. Luo, and Y. Fang, "Machine learning approaches for the prediction of postoperative complication risk in liver resection patients," *BMC medical informatics and decision making*, vol. 21, no. 1, p. 371, 2021.
- [4] T. Li, Y. Yang, J. Huang, R. Chen, Y. Wu, Z. Li, G. Lin, H. Liu, and M. Wu, "Machine learning to predict post-operative acute kidney injury stage 3 after heart transplantation," *BMC Cardiovascular Disorders*, vol. 22, no. 1, p. 288, 2022.
- [5] A. Heffernan, R. Ganguli, I. Sears, A. H. Stephen, and D. S. Heffernan, "Choice of machine learning models is important to predict post-operative infections in surgical patients," *Surgical Infections*, vol. 26, no. 7, pp. 520–529, 2025.
- [6] P. Yasin, Y. Yimit, X. Cai, A. Aimaiti, W. Sheng, M. Mamat, and M. Nijati, "Machine learning-enabled prediction of prolonged length of stay in hospital after surgery for tuberculosis spondylitis patients with unbalanced data: a novel approach using explainable artificial intelligence (xai)," *European journal of medical research*, vol. 29, no. 1, p. 383, 2024.
- [7] Y. Göktürk, S. K. Başarslan, Ş. Göktürk, H. Kocaman, and H. Yıldırım, "Prediction of postoperative haemorrhage after cerebral tumour surgery using machine learning algorithms," *BMC Medical Informatics and Decision Making*, vol. 25, no. 1, pp. 1–15, 2025.
- [8] R. Nopour, "Prediction of one-year recurrence among breast cancer patients undergone surgery using artificial intelligence-based algorithms: a retrospective study on prognostic factors," *BMC cancer*, vol. 25, no. 1, p. 940, 2025.
- [9] P. Mishra, *Explainable AI Recipes - Implement Solutions to Model Explainability and Interpretability with Python*. Apress Berkeley, CA, 2023.
- [10] M. Munn and D. Pitman, *Explainable AI for Practitioners: Designing and Implementing Explainable ML Solutions*. O'Reilly Media, 2022.
- [11] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "Smote: synthetic minority over-sampling technique," *J. Artif. Int. Res.*, vol. 16, no. 1, p. 321–357, Jun. 2002.
- [12] S. Chakraborty, "Topsis and modified topsis: A comparative analysis," *Decision Analytics Journal*, vol. 2, p. 100021, 2022. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S277266222100014X>
- [13] S. Brown, R. Tauler, and B. Walczak, *Comprehensive Chemometrics-Chemical and Biochemical Data Analysis, 2nd Eds*. Elsevier, 2020.
- [14] W. Zhou, C. Peng, P. Li, Y. Li, X. Liao, Z. Wang, M. Wang, Y. Xiao, H. Su, and H. Shi, "Machine learning-based prediction of short-term outcomes in aneurysmal subarachnoid hemorrhage: a multicenter study integrating clinical and inflammatory indicators," *BMC medicine*, vol. 24, no. 1, p. 7, 2026.
- [15] S. Summers and L. Woolery, "Post-Operative Patient," UCI Machine Learning Repository, 1991, DOI: <https://doi.org/10.24432/C5DG6Q>.
- [16] A. Tharwat, "Classification assessment methods," *Applied Computing and Informatics*, vol. 17, no. 1, pp. 168–192, 2018.
- [17] O. Rainio, J. Teuhio, and R. Klén, "Evaluation metrics and statistical tests for machine learning," *Scientific Reports*, vol. 14, p. 6086, 2024.
- [18] J. Bergstra and Y. Bengio, "Random search for hyper-parameter optimization," *Journal of Machine Learning Research*, vol. 13, no. 10, pp. 281–305, 2012.